

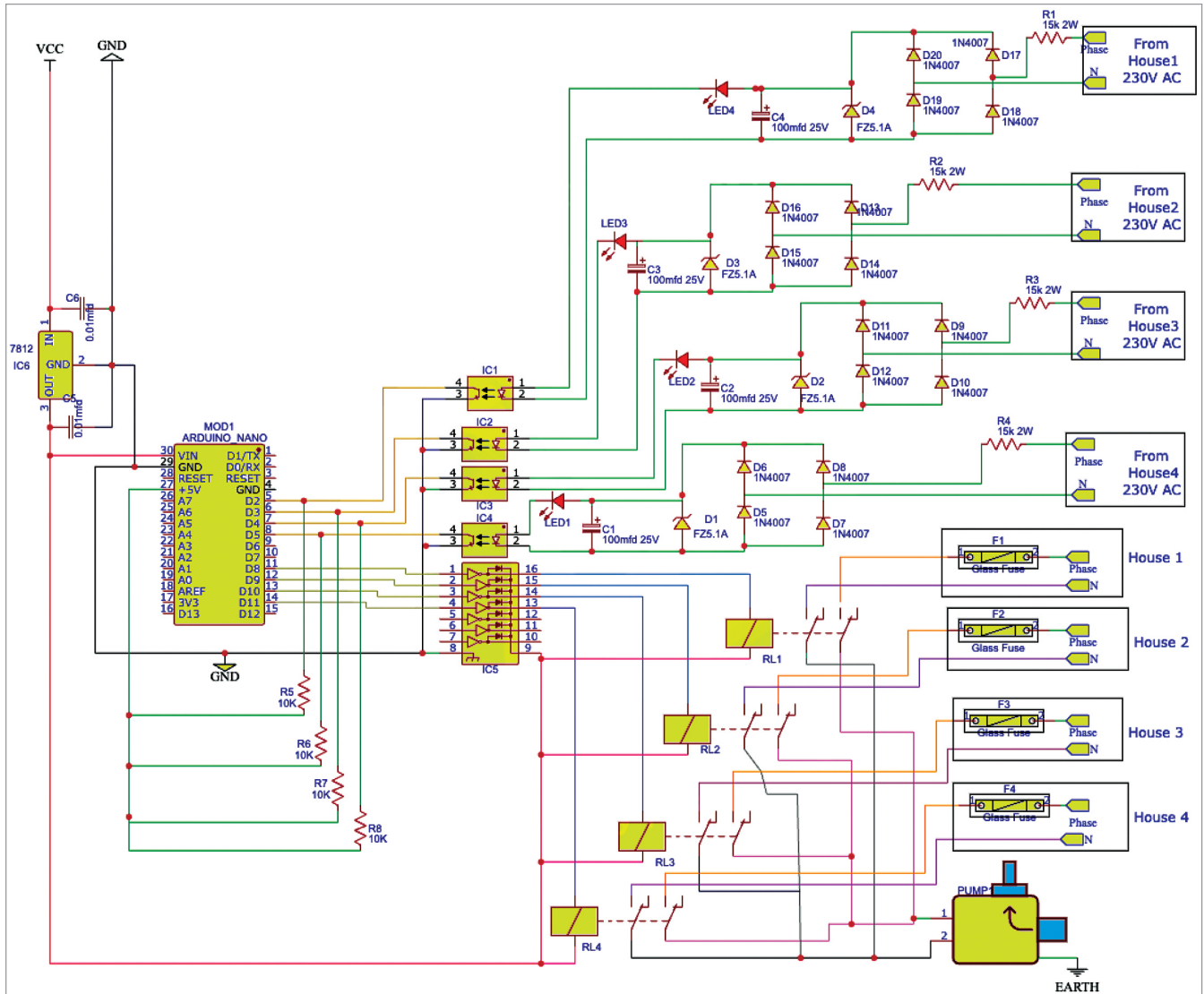


Fig. 2: Circuit diagram

an output terminal block for motor cable connections.

The control circuit is built around the Arduino Nano (Board1) and IC1 through IC6. It uses a 12V, 1A voltage regulator to provide 12V DC regulated power supply to the Arduino Nano board, relays (RL1-RL4), and optocouplers (IC1-IC4). The optocouplers sense power supply availability and motor switch status in all the houses, interfacing with the Arduino Nano board and relay driver IC ULN2003. The Arduino Nano board provides 5V power supply, which is available at its

5V terminal for the control circuit's inputs and outputs.

The Arduino Nano board receives input from a house's motor on/off switch through optocouplers (230V AC to 5V). The optocoupler IC PC817 provides electrical isolation, ensuring safe operation. When an optocoupler receives 230V AC supply, it switches the voltage from 5V to zero at the connected Arduino Nano pins (D2 through D5). This means that a logic high at an Arduino input is interpreted as the motor switch being off, and a logic

low as the motor switch being on.

The Arduino Nano board operates according to the embedded program. This program is designed to handle the demands of up to four houses sharing a single motor. However, it can be adjusted for just two or three houses as well. For more than four houses, both the program and hardware require modification.

The houses are prioritised based on their door numbers, ascending in order. For example, if there are four houses with door numbers 10, 11, 12, and 13, house number 10 receives the